

9. A police car is travelling at a constant speed of  $31.0 \text{ m s}^{-1}$  towards a stationary observer. The siren on the car emits a sound with a frequency of  $820 \text{ Hz}$ .

The speed of sound in air is  $340 \text{ m s}^{-1}$ .

The frequency of the sound heard by the observer is:

- A     $745 \text{ Hz}$
- B     $751 \text{ Hz}$
- C     $820 \text{ Hz}$
- D     $895 \text{ Hz}$
- E     $902 \text{ Hz}$ .